Fourier-Sector Decomposition, Hermite Diagonalization, and the Self-Dual Core

Gaussian is the vacuum. Hermite modes are the excitations. Fourier only rotates their phase.

1. Fourier transform and ladder operators

We work with the Fourier transform in the convention

$$(\mathcal{F}f)(\xi) := \int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx.$$

On the Schwartz space $\mathcal{S}(\mathbb{R})$, define

$$D_x := \frac{d}{dx}, \qquad M_x[f](x) := xf(x),$$

and the ladder operators

$$A := \frac{1}{2\sqrt{\pi}}D_x + \sqrt{\pi}M_x, \qquad A^\dagger := -\frac{1}{2\sqrt{\pi}}D_x + \sqrt{\pi}M_x.$$

Lemma 1 (Basic Fourier identities)**.** *For every $f \in \mathcal{S}(\mathbb{R})$,*

$$\mathcal{F}(D_x f)(\xi) = 2\pi i\,\xi\,\widehat{f}(\xi), \qquad \mathcal{F}(M_x f)(\xi) = \frac{i}{2\pi}D_\xi \widehat{f}(\xi),$$

where $\widehat{f} = \mathcal{F}f$.

Proof. For the derivative identity, integrate by parts:

$$\mathcal{F}(D_x f)(\xi) = \int_{\mathbb{R}} f'(x)e^{-2\pi i x\xi}\,dx = \Big[f(x)e^{-2\pi i x\xi}\Big]_{-\infty}^{+\infty} + 2\pi i\xi \int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx.$$

Since $f \in \mathcal{S}(\mathbb{R})$, the boundary term vanishes, hence

$$\mathcal{F}(D_x f)(\xi) = 2\pi i\,\xi\,\widehat{f}(\xi).$$

For multiplication by x, differentiate under the integral sign:

$$D_\xi \widehat{f}(\xi) = \frac{d}{d\xi}\int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx = \int_{\mathbb{R}} f(x)(-2\pi i x)e^{-2\pi i x\xi}\,dx.$$

Thus

$$\int_{\mathbb{R}} xf(x)e^{-2\pi i x\xi}\,dx = \frac{i}{2\pi}D_\xi \widehat{f}(\xi),$$

which is exactly

$$\mathcal{F}(M_x f)(\xi) = \frac{i}{2\pi}D_\xi \widehat{f}(\xi).$$

□

Lemma 2 (Commutator relation)**.** *On $\mathcal{S}(\mathbb{R})$,*

$$[A, A^\dagger] := AA^\dagger - A^\dagger A = I.$$

Proof. Since $[D_x, M_x] = I$, we compute

$$AA^\dagger = \left(\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right)\left(-\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right),$$

$$A^\dagger A = \left(-\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right)\left(\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right).$$

Subtracting, the diagonal terms cancel and only the mixed commutator survives:

$$[A, A^\dagger] = \frac{1}{2}[D_x, M_x] + \frac{1}{2}[D_x, M_x] = [D_x, M_x] = I.$$

□

2. Vacuum and intertwining

Define

$$h_0(x) := 2^{1/4} e^{-\pi x^2}.$$

Lemma 3 (Gaussian vacuum)**.** *The function h_0 satisfies*

$$Ah_0 = 0.$$

Moreover,

$$\mathcal{F}h_0 = h_0.$$

Proof. Differentiate:

$$\frac{d}{dx} h_0(x) = -2\pi x\, h_0(x).$$

Hence

$$Ah_0 = \frac{1}{2\sqrt{\pi}}(-2\pi x h_0) + \sqrt{\pi} x h_0 = -\sqrt{\pi} x h_0 + \sqrt{\pi} x h_0 = 0.$$

The identity $\mathcal{F}h_0 = h_0$ is the standard Gaussian Fourier identity

$$\mathcal{F}(e^{-\pi x^2})(\xi) = e^{-\pi \xi^2}.$$

Multiplying by the same normalization factor $2^{1/4}$ preserves the equality. □

Proposition 1 (Fourier intertwines ladder operators)**.** *On $\mathcal{S}(\mathbb{R})$,*

$$\mathcal{F}A = i\, A\, \mathcal{F}, \qquad \mathcal{F}A^\dagger = -i\, A^\dagger\, \mathcal{F}.$$

Proof. Using the previous Fourier identities,

$$\mathcal{F}(Af) = \frac{1}{2\sqrt{\pi}} \mathcal{F}(D_x f) + \sqrt{\pi}\, \mathcal{F}(M_x f) = \frac{1}{2\sqrt{\pi}}(2\pi i \xi \widehat{f}) + \sqrt{\pi}\left(\frac{i}{2\pi} D_\xi \widehat{f}\right).$$

This becomes

$$\mathcal{F}(Af) = i\left(\sqrt{\pi}\, \xi\, \widehat{f} + \frac{1}{2\sqrt{\pi}} D_\xi \widehat{f}\right) = i\, A(\widehat{f}),$$

so

$$\mathcal{F}A = i\, A\, \mathcal{F}.$$

Similarly,

$$\mathcal{F}(A^\dagger f) = -\frac{1}{2\sqrt{\pi}}\mathcal{F}(D_x f) + \sqrt{\pi}\,\mathcal{F}(M_x f) = -\frac{1}{2\sqrt{\pi}}(2\pi i\xi\widehat{f}) + \sqrt{\pi}\left(\frac{i}{2\pi}D_\xi\widehat{f}\right),$$

hence

$$\mathcal{F}(A^\dagger f) = -i\left(-\frac{1}{2\sqrt{\pi}}D_\xi\widehat{f} + \sqrt{\pi}\,\xi\,\widehat{f}\right) = -i\,A^\dagger(\widehat{f}).$$

Therefore

$$\mathcal{F}A^\dagger = -i\,A^\dagger\,\mathcal{F}.$$

□

3. Hermite modes

For $n \geq 0$, define

$$h_n := \frac{1}{\sqrt{n!}}(A^\dagger)^n h_0.$$

Lemma 4 (Commutator with higher powers)**.** *For every integer $n \geq 1$,*

$$[A, (A^\dagger)^n] = n(A^\dagger)^{n-1}.$$

Proof. We argue by induction on n.

For $n = 1$, this is exactly $[A, A^\dagger] = I$.

Assume

$$[A, (A^\dagger)^n] = n(A^\dagger)^{n-1}.$$

Then

$$[A, (A^\dagger)^{n+1}] = [A, (A^\dagger)^n A^\dagger] = [A, (A^\dagger)^n]A^\dagger + (A^\dagger)^n[A, A^\dagger].$$

By the induction hypothesis and $[A, A^\dagger] = I$,

$$[A, (A^\dagger)^{n+1}] = n(A^\dagger)^{n-1}A^\dagger + (A^\dagger)^n = n(A^\dagger)^n + (A^\dagger)^n = (n+1)(A^\dagger)^n.$$

This proves the claim. □

Proposition 2 (Ladder action)**.** *For $n \geq 0$,*

$$A^\dagger h_n = \sqrt{n+1}\,h_{n+1},$$

and for $n \geq 1$,

$$Ah_n = \sqrt{n}\,h_{n-1}.$$

Proof. The raising relation is immediate:

$$A^\dagger h_n = A^\dagger\left(\frac{1}{\sqrt{n!}}(A^\dagger)^n h_0\right) = \frac{1}{\sqrt{n!}}(A^\dagger)^{n+1}h_0 = \sqrt{n+1}\,h_{n+1}.$$

For the lowering relation, use the commutator lemma:

$$Ah_n = \frac{1}{\sqrt{n!}}A(A^\dagger)^n h_0 = \frac{1}{\sqrt{n!}}\Big((A^\dagger)^n A + [A, (A^\dagger)^n]\Big)h_0.$$

Since $Ah_0 = 0$, the first term vanishes, so

$$Ah_n = \frac{1}{\sqrt{n!}}\,n(A^\dagger)^{n-1}h_0 = \sqrt{n}\,\frac{1}{\sqrt{(n-1)!}}(A^\dagger)^{n-1}h_0 = \sqrt{n}\,h_{n-1}.$$

□

Proposition 3 (Number operator)**.** *Define the number operator*

$$N := A^\dagger A.$$

Then

$$Nh_n = nh_n \qquad (n \geq 0).$$

Proof. For $n = 0$,

$$Nh_0 = A^\dagger(Ah_0) = 0.$$

So the formula holds for $n = 0$.

For $n \geq 1$, use the lowering relation:

$$Nh_n = A^\dagger(Ah_n) = A^\dagger(\sqrt{n}\, h_{n-1}) = \sqrt{n}\, A^\dagger h_{n-1} = \sqrt{n}\,\sqrt{n}\, h_n = nh_n.$$

□

Theorem 1 (Hermite modes are Fourier eigenfunctions)**.** *For every $n \geq 0$,*

$$\mathcal{F}h_n = (-i)^n h_n.$$

Proof. By definition,

$$h_n = \frac{1}{\sqrt{n!}}(A^\dagger)^n h_0.$$

Apply $\mathcal{F}$:

$$\mathcal{F}h_n = \frac{1}{\sqrt{n!}}\mathcal{F}\big((A^\dagger)^n h_0\big).$$

Using $\mathcal{F}A^\dagger = -i\, A^\dagger\, \mathcal{F}$ repeatedly,

$$\mathcal{F}\big((A^\dagger)^n h_0\big) = (-i)^n (A^\dagger)^n \mathcal{F}h_0.$$

Since $\mathcal{F}h_0 = h_0$,

$$\mathcal{F}h_n = \frac{(-i)^n}{\sqrt{n!}}(A^\dagger)^n h_0 = (-i)^n h_n.$$

□

4. General functions and diagonalization

At this point we use the classical completeness theorem for Hermite functions.

Theorem 2 (Classical Hermite completeness)**.** *The family $\{h_n\}_{n\geq 0}$ is an orthonormal basis of $L^2(\mathbb{R})$.*

Remark 1. *This is the only standard external input in the argument. It follows from the spectral theory of the harmonic oscillator.*

Theorem 3 (Fourier is diagonal in the Hermite basis)**.** *For every $f \in L^2(\mathbb{R})$, there exist coefficients $c_n = \langle f, h_n\rangle$ such that*

$$f = \sum_{n=0}^{\infty} c_n h_n \qquad \text{in } L^2(\mathbb{R}),$$

and

$$\mathcal{F}f = \sum_{n=0}^{\infty} (-i)^n c_n h_n.$$

Equivalently,

$$\boxed{f = \sum_{n\geq 0} c_n h_n \quad \Longrightarrow \quad \mathcal{F}f = \sum_{n\geq 0} (-i)^n c_n h_n.}$$

Proof. By Hermite completeness,

$$f = \sum_{n=0}^{\infty} c_n h_n \qquad \text{with } c_n = \langle f, h_n \rangle.$$

Since $\mathcal{F}$ is unitary on $L^2(\mathbb{R})$, it may be applied termwise in the L^2-sense:

$$\mathcal{F} f = \sum_{n=0}^{\infty} c_n \, \mathcal{F} h_n.$$

Using $\mathcal{F} h_n = (-i)^n h_n$,

$$\mathcal{F} f = \sum_{n=0}^{\infty} (-i)^n c_n h_n.$$

□

Corollary 1 (Operator form)**.** *On $L^2(\mathbb{R})$,*

$$\boxed{\mathcal{F} = e^{-i\pi N/2}.}$$

Proof. For each n,

$$e^{-i\pi N/2} h_n = e^{-i\pi n/2} h_n = (-i)^n h_n = \mathcal{F} h_n.$$

So the two operators agree on the orthonormal basis $\{h_n\}_{n \geq 0}$, hence on all of $L^2(\mathbb{R})$. □

5. Fourier-sector decomposition

Since

$$\mathcal{F}^4 = I,$$

the spectrum of $\mathcal{F}$ is contained in

$$\{1, -i, -1, i\}.$$

For each $\lambda \in \{1, -i, -1, i\}$, define

$$P_\lambda := \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^m.$$

Theorem 4 (Fourier-sector projectors)**.** *For every $\lambda \in \{1, -i, -1, i\}$,*

$$\mathcal{F}(P_\lambda f) = \lambda \, P_\lambda f \qquad \forall f \in L^2(\mathbb{R}).$$

Moreover, each P_λ is the spectral projector onto the λ-eigenspace of $\mathcal{F}$, and every $f \in L^2(\mathbb{R})$ decomposes as

$$f = P_1 f + P_{-i} f + P_{-1} f + P_i f.$$

Proof. By definition,

$$P_\lambda = \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^m.$$

Apply $\mathcal{F}$:

$$\mathcal{F} P_\lambda = \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^{m+1}.$$

Reindex with $k = m + 1$. Since $\mathcal{F}^4 = I$ and $\lambda^4 = 1$, this becomes

$$\mathcal{F}P_\lambda = \lambda \frac{1}{4} \sum_{k=0}^{3} \lambda^{-k} \mathcal{F}^k = \lambda P_\lambda.$$

Hence

$$\mathcal{F}(P_\lambda f) = \lambda P_\lambda f.$$

Thus $P_\lambda f$ lies in the λ-eigenspace.

Because the polynomial

$$z^4 - 1 = (z-1)(z+i)(z+1)(z-i)$$

splits into distinct linear factors, the projectors P_λ resolve the identity. Therefore every $f \in L^2(\mathbb{R})$ decomposes into the four Fourier sectors:

$$f = P_1 f + P_{-i} f + P_{-1} f + P_i f.$$

□

6. The self-dual core

The most important sector is $\lambda = 1$. It is the self-dual sector:

$$P_1 = \frac{1}{4}\left(I + \mathcal{F} + \mathcal{F}^2 + \mathcal{F}^3\right).$$

Thus for every $f \in L^2(\mathbb{R})$,

$$\boxed{P_1 f = \frac{1}{4}\left(f + \mathcal{F}f + \mathcal{F}^2 f + \mathcal{F}^3 f\right), \qquad \mathcal{F}(P_1 f) = P_1 f.}$$

Corollary 2 (Self-dual core)**.** *If*

$$f = \sum_{n=0}^{\infty} c_n h_n,$$

then

$$P_1 f = \sum_{\substack{n \geq 0 \\ n \equiv 0 \,(\mathrm{mod}\, 4)}} c_n h_n.$$

So $P_1 f$ is exactly the part of f made of Hermite modes whose Fourier phase is 1.

Proof. Since

$$\mathcal{F}h_n = (-i)^n h_n,$$

we have

$$P_1 h_n = \frac{1}{4} \sum_{m=0}^{3} \mathcal{F}^m h_n = \frac{1}{4} \sum_{m=0}^{3} (-i)^{mn} h_n.$$

If $(-i)^n = 1$, equivalently $n \equiv 0 \pmod 4$, then the sum equals 4, so $P_1 h_n = h_n$. Otherwise the four roots sum to 0, so $P_1 h_n = 0$.

Applying this termwise to

$$f = \sum_{n=0}^{\infty} c_n h_n$$

gives

$$P_1 f = \sum_{n \equiv 0 \,(\mathrm{mod}\, 4)} c_n h_n.$$

□

Every signal has a self-dual core.

Fourier does not spare one lucky function. It spares an entire operator-generated family.